



LINEAR ALGEBRA AND ITS APPLICATIONS

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CLASS-8

SYMMETRIC MATRICES AND DIAGONALIZATION OF A MATRIX

Statement: If A is a square matrix of order ' n ' has ' n ' linearly independent vectors, then a matrix ' S ' can be found such that $S^{-1}AS$ is a diagonal matrix.

Proof:

Let A be a square matrix of order 3. Let λ_1, λ_2 and λ_3 be its Eigen value and $X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ and

$X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$ be the corresponding Eigen vectors.

Let the square matrix $S = [x_1 \ x_2 \ x_3]$ i.e.

$$S = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$$

Multiplying by A ,

$$AS = A[x_1 \ x_2 \ x_3] = [Ax_1, Ax_2, Ax_3]$$

$$AS = [\lambda_1 x_1, \lambda_2 x_2, \lambda_3 x_3]$$

$$AS = \begin{bmatrix} \lambda_1 x_1 & \lambda_2 x_2 & \lambda_3 x_3 \\ \lambda_1 y_1 & \lambda_2 y_2 & \lambda_3 y_3 \\ \lambda_1 z_1 & \lambda_2 z_2 & \lambda_3 z_3 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = SD$$

$\therefore AS = SD$ where D is the diagonal matrix $D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$

$$\therefore AS = SD$$

Multiplying both sides by S^{-1}

$$\boxed{S^{-1}AS = D} \quad \text{or} \quad S^{-1}AS = \Lambda$$

$$\text{or} \quad \boxed{A = SDS^{-1}}$$

S is invertible because its columns (the eigen vectors) were assumed to be independent.

NOTE:

1. Any matrix with distinct Eigen Values can be diagonalise.
 2. Not all matrices has 'n' linearly independent Eigen vectors. Therefore, all the matrices can't be diagonalized.
- Eg: $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, here $\lambda_1 = \lambda_2 = 0$. There is only one independent vector. Hence we cannot construct S.

3. If Eigen vectors $x_1, x_2, \dots, x_k$ correspond to distinct Eigen values $\lambda_1, \lambda_2, \dots, \lambda_k$ then those Eigen vectors are linearly independent.
4. Eigen vector matrix is not unique since if x is an Eigen vector corresponding to λ then kx is also an Eigen vector.

Problem

1. Factor the matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ into $S \Lambda S^{-1}$ &

also find $S \Lambda S^{-1}$.

Ans: Consider $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = 0 \Rightarrow \begin{aligned} (1-\lambda)^2 - 1 &= 0 \\ 1 + \lambda^2 - 2\lambda - 1 &= 0 \Rightarrow \lambda^2 - 2\lambda = 0 \end{aligned}$$

$$\lambda(\lambda - 2) = 0 \Rightarrow \lambda_1 = 0 \text{ \& } \lambda_2 = 2$$

$$\therefore \Lambda = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

Now to find Eigen vectors:-

$$\text{Consider } (A - \lambda I)x = 0$$

$$\Rightarrow (1 - \lambda)x + y = 0$$

$$x + (1 - \lambda)y = 0$$

$$\text{Case 1:- When } \lambda = 2 \Rightarrow \left. \begin{array}{l} -x + y = 0 \\ x - y = 0 \end{array} \right\} \Rightarrow x = y$$

$\Rightarrow$ 1 equations with 2 unknowns $\therefore y$ be free variable

$$\therefore x = y = k, \text{ let } k = 1$$

$$\text{Eigen vector } x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Case 2:- When $\lambda = 0$

$$\left. \begin{array}{l} x+y=0 \\ x+y=0 \end{array} \right\} \Rightarrow x=-y$$

$\Rightarrow$ 2 unknowns so one free variable

$$x=k \Rightarrow y=-k \text{ or } x=-k \Rightarrow y=k$$

$$\therefore x_2 = \begin{bmatrix} -k \\ k \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\therefore S = [x_1 \ x_2] = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}, N = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}, S^{-1} = \frac{1}{2} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\therefore A = S N S^{-1} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Here verified.

To find $SA S^{-1}$

$$\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \Lambda$$



THANK YOU

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